Interpretable Prediction of Reported Older Person's Grant Non-Receipt Among Apparently Eligible Adults in KwaZulu-Natal

Oluhle Jawe
Department of Information Technology
Durban University of Technology
Durban, South Africa

Abstract— This study proposes an interpretable machine learning artefact for examining reported non-receipt of the Older Person's Grant (OPG) among adults aged 60 years and older in KwaZulu-Natal. The Living Conditions Survey (LCS) 2014/2015 will be used to construct a study population that meets eligibility conditions observable in the survey. This population will be described as apparently eligible because the data cannot establish every legal requirement. A structured literature review informs the outcome definition, candidate predictors and evaluation approach. The proposed analysis will combine survey-weighted descriptions with a small logistic regression model evaluated using household-grouped cross-validation. A Python workflow and Streamlit research interface will connect data preparation, eligibility screening, model evaluation and reporting. The study aims to show how screening assumptions affect estimates of non-receipt and whether the selected characteristics provide useful predictive information. This paper presents the rationale and proposed methods; empirical findings will depend on subsequent analysis.

Keywords— Older Person's Grant; reported non-receipt; KwaZulu-Natal; Living Conditions Survey; logistic regression; research artefact.

I. Introduction

The Older Person's Grant forms part of South Africa's social assistance system [1]. Studying its reach requires a distinction between being old enough for the programme, meeting its eligibility requirements and reporting receipt. An older person who reports no grant may have resources above the means test, may not have applied, or may be awaiting a decision. A household survey may also contain incomplete information. Reported non-receipt therefore requires a clearly defined study population and careful interpretation.

Previous research has examined differences between pension recipients and non-recipients in rural South Africa [2]. However, findings based on age eligibility or a particular locality cannot automatically describe apparently eligible older adults across KwaZulu-Natal. The way income, marital circumstances and missing information are handled may change who enters the analysis and, consequently, the estimated extent of non-receipt.

The LCS provides person, household and income information that can support this investigation [3], [4]. It also distinguishes OPG receipt from other social grants [5]. These features allow a person-level analysis that considers household circumstances while maintaining a grant-specific outcome. The survey does not provide a complete legal assessment, application history or administrative payment record. Throughout this paper, apparently eligible refers to adults who pass the study's observable screening rules, rather than confirmed legal eligibility.

The ICT component will be a reproducible research artefact that applies these rules, produces descriptive outputs and evaluates a probability model. Its intended value is to make the analysis understandable and repeatable. It will estimate reported non-receipt within a historical survey population, rather than predict future grant decisions.

II. Aim and Research Questions

The aim is to develop and evaluate an interpretable computational artefact for analysing reported OPG non-receipt among apparently eligible older adults in KwaZulu-Natal using the LCS 2014/2015.

The study will address three research questions:

RQ1: What proportion of the study population reports OPG non-receipt, and how does this estimate change under alternative eligibility assumptions?

RQ2: How well does a small logistic regression model distinguish recipients from non-recipients and estimate non-receipt probabilities in held-out households?

RQ3: Can the proposed artefact reproduce the analysis and clearly explain the screening decisions and model outputs?

RQ1 addresses the substantive question of non-receipt and RQ2 its predictive assessment; RQ3 is a design-evaluation question concerning the artefact's transparency and reproducibility.

The objectives are to construct a documented person-level study population; describe receipt and non-receipt using appropriate survey weights; develop and internally validate a parsimonious model; and evaluate the artefact's correctness, reproducibility and clarity. Together, these objectives connect the substantive question of grant access with a focused system development task.

III. Literature Review

A. Review approach

A structured literature review with thematic narrative synthesis was used to examine grant access, eligibility measurement and prediction. Searches covered Scopus, Dimensions, IEEE Xplore, SABINET/African Journals Online, Google Scholar and relevant Statistics South Africa and DataFirst resources. Search concepts included social assistance, eligibility, take-up, non-receipt, exclusion and prediction. Sources were considered for their relevance to

the research question, population, outcome and methodological contribution.

The searches identified 511 records. After removing 136 duplicates, 375 unique records underwent title and abstract screening, which excluded 222 records. The remaining 153 records were assessed at full text; 98 were excluded and 55 sources were retained in the broader social-grant review. These counts describe the selection process for that review, rather than 55 OPG-specific studies. The present synthesis draws on directly relevant OPG evidence and selected findings from other grants that inform eligibility measurement, access mechanisms and predictive methods.

The synthesis compares studies by programme, setting, unit of analysis, outcome definition, findings and limitations. Direct OPG evidence informs the substantive question. Studies of other grants are used where they clarify access processes or methodological choices. Their findings are not treated as estimates of OPG non-receipt. A narrative approach is appropriate because the studies examine different populations and stages of grant access, making a single pooled estimate unsuitable.

B. OPG receipt and the KwaZulu-Natal context

Ralston et al. [2] examine pension receipt among age-eligible adults in rural Agincourt. Their person-level approach is relevant to identifying characteristics associated with receipt. However, age eligibility does not establish whether an individual satisfies all resource requirements, and the rural setting limits direct transfer to a province-wide KwaZulu-Natal population. The study provides a precedent for the proposed outcome while showing why the eligibility denominator needs to be specified.

Godfrey et al. [6] examine grant receipt in rural and peri-urban communities in Msunduzi, KwaZulu-Natal. Their outcomes concern the proportions of age-eligible household members receiving grants. Associations differ between the old-age and child-grant models, suggesting that household characteristics do not have a uniform relationship with receipt across programmes. The study is geographically relevant, but a household proportion is different from an individual's receipt status. A household-level association cannot be assumed to describe the probability that a particular older adult receives OPG.

Raniga and Simpson [7] examine the old-age pension in the livelihoods of households in Bhambayi. Alloush et al. [8] consider social protection through the OPG during a later crisis. These studies help explain why pension access matters for households. Their emphasis on livelihoods and programme consequences differs from estimating non-receipt among apparently eligible individuals. Taken together, the evidence supports attention to both the individual beneficiary and the household context, without treating the two units as interchangeable.

C. Eligibility and the meaning of non-receipt

A recurring issue is that contact with a grant programme, application, receipt and continued payment are different outcomes. Case et al. [9] examine grant contact in rural KwaZulu-Natal, including stages such as enquiry and application. Their findings therefore extend beyond successful payment alone. Zembe-Mkabile et al. [10] distinguish application from receipt among mothers of young infants in Umlazi. Some applicants were still awaiting processing at observation. These studies show that non-receipt at one point in time need not mean rejection or permanent exclusion.

The implications extend to eligibility measurement. Gazeaud [11] examines the sensitivity of proxy means testing to measurement error in Tanzania. Although that targeting system differs from the proposed income screen, the study illustrates how resource measurement can affect classification. It supports testing the sensitivity of the study population to income assumptions, without establishing South African eligibility rules or the size of any error in the LCS.

For the present study, eligibility screening and receipt classification will therefore be separate steps. The outcome will refer specifically to reported OPG receipt. Current receipt and income received over an earlier reference period will be compared rather than assumed to measure the same event. Records with unresolved information will remain identifiable, since treating unknown values as non-receipt could overstate the outcome.

D. Administrative and geographic access

Administrative-burden theory distinguishes the costs of learning about a programme, complying with requirements and dealing with the psychological demands of access [12]. South African studies describe how such costs may arise at different stages. Heinrich and Brill [13] examine payment interruption and disconnection among Child Support Grant (CSG) beneficiaries. Their findings concern continuity among people who entered the programme, rather than all eligible people who never received it.

Gibbs et al. [14] describe documentary, financial and interpersonal barriers experienced by young women in KwaZulu-Natal. Majola and Maharaj [15] discuss information gaps, queues, system problems and repeated application costs among young mothers who ultimately received CSG. Together, these qualitative studies suggest that difficult access can coexist with eventual receipt. They help explain possible mechanisms but do not establish their prevalence among older adults or their effects on OPG non-receipt.

Granlund [16] highlights travel and waiting costs associated with payment arrangements in the rural Eastern Cape. Vujovic et al. [17] examine digital and financial access in relation to social protection. These studies suggest that geography and access conditions may be relevant, but their settings, periods and outcomes differ from the proposed analysis. Settlement and welfare-office access will therefore be examined as possible contextual predictors. Their associations will not, by themselves, demonstrate administrative failure or explain why a person did not receive OPG.

E. Predictors and model evaluation

The literature supports examining demographic, educational, household and geographic circumstances, while offering limited grounds for assuming a common direction of association [2], [6], [9]. Income and consumption may partly reflect receipt of a grant. Including direct receipt information or its derivatives as predictors could make a model appear useful because it indirectly observes the outcome. The proposed model will therefore separate the

variables used for resource screening from the variables used for prediction.

Mabala [18] uses logistic modelling to study Social Relief of Distress Grant outcomes among young applicants. This provides a methodological comparison, but applicant-based findings for a different grant and age group cannot be transferred directly to older adults. The relevant lesson is to define the population and event clearly before interpreting a model's coefficients or predictive performance.

Methodological research also supports evaluating more than classification accuracy. Calibration concerns agreement between predicted probabilities and observed outcomes [19]. Precision-recall assessment is useful when the event of interest is uncommon [20]. The number of model parameters should be considered alongside sample size and event frequency [21]. Grouped validation helps address dependence between related observations [22], while TRIPOD provides guidance for reporting model development and evaluation transparently [23]. These principles favour a small, interpretable baseline and explicit assessment of uncertainty.

F. Synthesis and research gap

Three linked issues emerge from the reviewed evidence. First, the definition of eligibility determines the population in which non-receipt is measured. Second, studies of application, receipt and payment interruption explain different stages of access. Third, plausible predictors require empirical evaluation, and predictive associations should not be interpreted as causal explanations.

The proposed contribution is to bring transparent eligibility screening, survey-weighted description and internally validated prediction into one reproducible analysis of older adults in KwaZulu-Natal. This is a specific opportunity identified within the reviewed literature, rather than a claim that OPG receipt has not previously been studied. The artefact will make it possible to examine how alternative screening assumptions affect both the descriptive estimates and the model's assessment.

IV. Proposed Data and Eligibility Methods

A. Study design and data preparation

The study will use a quantitative, cross-sectional secondary analysis of LCS 2014/2015. The target survey population is adults aged 60 years and older living in sampled KwaZulu-Natal households. The survey's household coverage does not represent institutionalised older adults, and its historical reference period limits conclusions about present-day access [3].

Person records will be linked to household information and person-attributable income using the documented household and person identifiers. Income records will be classified using the survey's COICOP definitions and aggregated to the individual before linkage [3], [4]. Checks will identify duplicate person keys, unmatched records and unexpected changes in record counts. The released variables and questionnaire definitions will guide the interpretation of values and structural skips [5].

Figure 1 shows the proposed cohort construction. Screening will distinguish people who pass the observable rules, people with a known rule failure, and people whose status cannot be resolved. Reasons for exclusion or uncertainty will be retained. A final analysis record will require both a passing screen and a resolved OPG receipt response.

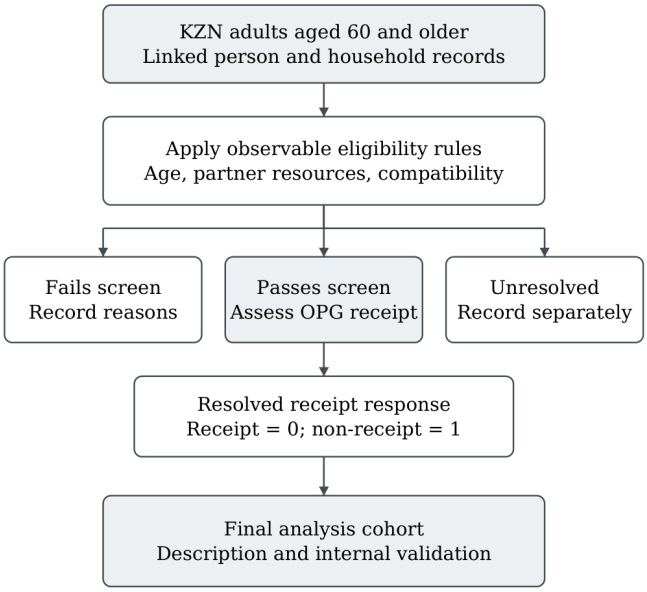


Fig. 1. Proposed construction of the analysis cohort. Unresolved receipt responses are accounted for separately and do not enter the primary model. The diagram specifies a process; it does not report sample counts.

B. Partner resources and the income screen

For married respondents, a resident spouse will be linked using the questionnaire's spouse identifier within the household. The study will check that the linked person exists and that the relationship is consistent. A missing or non-resident spouse with unavailable resources will produce an unresolved income screen rather than automatic use of a single-person threshold. Ambiguous partnership categories will be treated conservatively and examined through stated sensitivity scenarios.

The proposed resource proxy is annual nominal income attributable to the applicant and, where applicable, the spouse. It will use the survey's annualised values without annualising them again. The analytical definition will exclude recorded social-assistance income and will distinguish it from other income sources [4]. This is a research proxy: it may not capture all legal deductions, income definitions or assessable resources. Household totals will not be allocated arbitrarily to individual applicants.

An absent income record will not automatically be treated as zero. Zero income must be supported by the survey's release conventions or evidence that the relevant schedule is complete. Otherwise, resource status will remain unresolved. Similarly, asset acquisitions will not be treated as a complete stock of legally assessable assets. The absence of a full asset assessment is one reason the study uses the term apparently eligible.

C. Eligibility assumptions and outcome definition

The screening specification will be finalised before predictive performance is examined. Income thresholds, comparison boundaries and partnership treatment will be checked against official provisions relevant to the survey period. Any simplified threshold scenario will be explicitly identified as an analytical assumption. Rule versions will allow the same records to be reassessed under alternative

assumptions without silently changing the primary study population.

Reported receipt of an own disability or war-veterans grant will be flagged separately and excluded in the conservative primary screen, with an alternative scenario assessing the effect of retaining these records. Caregiver-linked child grants and grant-in-aid will not automatically be treated as incompatible pensions. Unresolved responses will be retained as uncertainty rather than assigned a definite status.

Within the screened population, the binary outcome will be Y = 1 for reported OPG non-receipt and Y = 0 for reported receipt. The general social-grant question and the OPG-specific response will be interpreted together, following the questionnaire's skip rules [5]. An explicit, consistent report of no social grant can support a non-receipt label. Contradictory responses and unexplained missing values will remain unresolved. Receiving CSG as a caregiver will not determine the OPG label.

Current reported receipt will be compared with preceding-year OPG income to identify disagreement between reference periods. Such disagreement will be examined in sensitivity analysis rather than automatically corrected in favour of one measure. Cohort reporting will include people, households, known screening failures, unresolved eligibility and unresolved outcomes, so that the denominator remains visible.

V. Proposed Statistical Analysis

A. Descriptive analysis

The unweighted non-receipt proportion will be the number of non-recipients divided by the number of people with resolved outcomes in the screened population. The weighted proportion will be calculated as $\Sigma\left(w_i Y_i\right) / \Sigma w_i$, using the released person weights. Recipient and non-recipient characteristics will be described with both unweighted counts and weighted summaries, keeping their different purposes clear.

The availability of strata and primary sampling unit information will determine how survey uncertainty can be estimated. Where possible, the analysis will retain the full survey design when estimating the KwaZulu-Natal older-adult domain. If the design information is incomplete, weighted point estimates will remain descriptive and any variance approximation will be identified. Unknown outcomes among screen-passing adults will also be assessed by assigning them first to receipt and then to non-receipt to show outcome-uncertainty bounds.

B. Predictor specification and feasibility

Table I presents the proposed primary predictors. The specification contains seven slope coefficients and an intercept. Age, sex and schooling represent individual characteristics, while settlement and welfare-office access represent geographic context. Marital circumstances inform eligibility screening but will not add another coefficient to the primary model. The model estimates associations within the screened population; it does not identify causes of non-receipt.

Age and sex are motivated by Ralston et al.'s comparison of pensioners and non-pensioners [2], while schooling reflects the educational and information-related dimensions of access discussed by Case et al. [9] and Majola and Maharaj [15]. Settlement represents the geographic context of access described in these studies [9], [15]. Welfare-office access is motivated more directly by Granlund's account of travel and waiting costs [16], with Vujovic et al. [17] providing complementary evidence on geographic and digital access conditions. These findings motivate candidate predictors, not assumed causal effects or directions of association.

TABLE I
PROPOSED PRIMARY PREDICTOR SPECIFICATION

Predictor	Proposed coding	df
Age	Linear, per 10 years above 60	1
Sex	Male and female categories	1
Schooling	Below Grade 7 versus Grade 7 or above	1
Settlement	Four documented categories	3
Welfare access	At least 5 km or unavailable versus nearer access	1
Total	Intercept additional	7

A preliminary metadata check confirms a realised national sample of 27,527 households and an 89.6% response rate in KwaZulu-Natal [3]; the size and non-receipt event count of the screened KZN population aged 60 and older will still be verified before model fitting.

Feasibility will be assessed using cohort size, non-recipient events, household counts and the distribution of categories across validation folds. Sample-size guidance [21] will inform this assessment but cannot guarantee adequacy for clustered survey data. If the planned model is unsupported, a simpler specification will require a documented revision. Stepwise selection based on statistical significance will not be used.

Missing predictor values will be handled within training folds using median imputation for continuous variables and modal imputation for categorical variables. Structural skips will be interpreted before imputation. Eligibility and outcome information will not be filled using this procedure. A complete-case sensitivity analysis will assess whether predictor missingness materially changes the study population or findings.

C. Model development and internal validation

The primary estimator will be an unweighted logistic regression model. It will estimate the probability of reported non-receipt from the seven specified predictor columns. Survey-weighted fitting will be a separate sensitivity analysis. No outcome oversampling or class balancing is planned. Convergence, separation and the support for each model term will be checked, and estimation failures will be reported.

Five-fold stratified group cross-validation will keep all members of a household in the same fold while seeking approximate outcome balance. All imputation and other fitted preparation steps will use training data only. Each person will receive a prediction from a model that did not use their household for training. A fold without both outcome classes will require review of the validation plan. A primary-sampling-unit grouping sensitivity will be

considered because neighbouring households may still share contextual information [22].

Evaluation will combine average precision with a calibration plot and calibration slope. Average precision will be interpreted alongside non-receipt prevalence [20]. ROC AUC, the Brier score and calibration-in-the-large will provide complementary assessments [19]. An intercept-only model fitted within the same folds will provide a simple reference for probability error. Primary metrics will be unweighted; weighted metrics will be reported separately as a sensitivity.

Uncertainty intervals for the pooled held-out predictions will be estimated by resampling households. These intervals describe uncertainty in the fixed prediction set and do not capture every source of variation from retraining the model. Invalid resamples and unstable calibration estimates will be reported. The analysis will assess probabilities rather than select a threshold for operational grant decisions.

D. Sensitivity analysis and model comparison

Sensitivity analyses will vary declared assumptions concerning thresholds, partner resources, grant compatibility, income definitions and receipt-period disagreement. Predictor sensitivities will include complete cases and replacing welfare access with partnered status at the same coefficient count. Each scenario will report its own denominator and event rate, since a change in predictive performance may partly reflect a change in the study population.

One exploratory random-forest comparison may be undertaken if the available events and validation design support it. It will use the same eligible observations, predictor information and outer household partitions. Any tuning will be restricted to training data using grouped inner validation. The comparison will remain secondary; improved performance will not be assumed, and limited sample support may justify retaining logistic regression alone.

VI. Proposed Computational Artefact

The artefact will comprise a Python analysis package and a Streamlit research review interface. Figure 2 shows how the components will connect. The preparation module will validate and link the survey files. A rule module will assign screening decisions and reasons. Descriptive and predictive modules will then produce their respective outputs from the defined cohort.

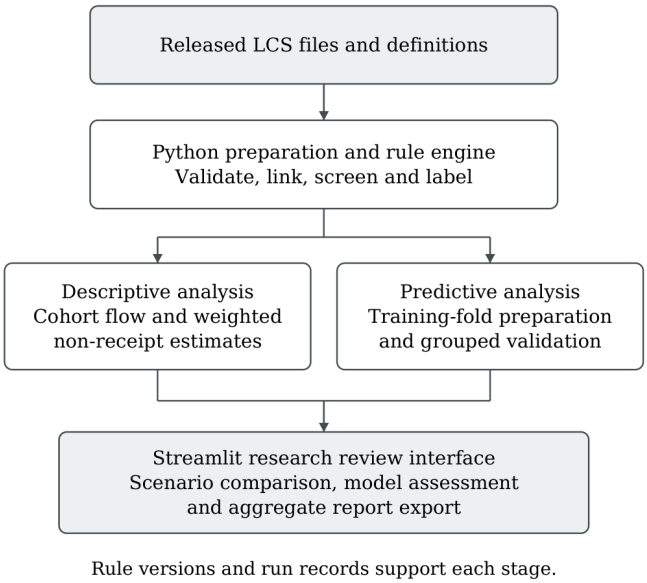


Fig. 2. Proposed artefact architecture. Both analysis modules use the screened cohort. Receipt labels define the model outcome and are excluded from predictor preparation.

The interface will allow a researcher to select a documented scenario, inspect aggregate cohort counts, compare non-receipt estimates and review model performance. Outputs will identify the survey period, rule version and model specification. Aggregate tables and figures will be exportable for reporting. A command-line workflow will support repeatable execution of the same analysis.

Reproducibility records will include input-file checksums, variable mappings, software versions, random seeds and validation partitions. The application will be deployed as a controlled research prototype, with synthetic records used for public demonstration. Access to the underlying survey files will remain subject to the data-use conditions.

A. Artefact evaluation

Correctness tests will use independently specified synthetic cases covering income boundaries, missing partner resources, unresolved receipt and caregiver-linked grants. Expected screening decisions will be compared with the system output. Linkage checks will verify that the process preserves the intended person-level unit. Predictor checks will reject direct grant-receipt information and derived labels.

Reproducibility will be assessed by repeating the workflow with the same inputs and configuration and comparing cohort membership, reason codes and model outputs within a stated numerical tolerance. Interface evaluation will use defined research tasks, such as locating an exclusion reason, identifying the active scenario and interpreting a calibration display. Task completion and identified misunderstandings will guide revision. These are proposed evaluation activities, not completed findings.

The minimum viable product (MVP) comprises the rule engine and screening logic, weighted descriptions, logistic model and household-grouped cross-validation needed for RQ1 and RQ2. For RQ3, recorded screening reasons, saved rule and model settings, validation partitions and a repeatable command-line workflow remain essential. The Streamlit interface, extended reproducibility automation and exploratory random-forest comparison are secondary and may be reduced or deferred if time is constrained, while retaining the core analysis and its evaluation.

VII. Ethics and Study Limitations

The study will follow the survey's access conditions and obtain the required DUT ethics determination for the proposed research. Person-level records, household identifiers and predictions will remain in controlled research storage. Reporting will follow applicable disclosure requirements, and public examples will use synthetic data. The interface will support research interpretation and will not issue official eligibility decisions or rank named applicants.

The main limitations are incomplete observation of legal eligibility, possible income and receipt measurement error, and a potentially small number of non-recipients. Cross-sectional associations cannot establish why non-receipt occurred or whether the selected factors preceded it. Survey-design constraints may limit uncertainty estimation, while internal validation cannot establish performance in a new province, period or administrative setting.

The expected contribution is a transparent analysis of how observable screening choices shape reported OPG non-receipt, supported by a reproducible ICT artefact. The study will assess whether a small model adds useful predictive information without assuming that it will perform well. Weak performance would still help establish the limits of prediction from the available survey characteristics.

References

[1] Republic of South Africa, Social Assistance Act 13 of 2004, Government Gazette 26446, Jun. 10, 2004. [Online]. Available: https://www.gov.za/documents/social-assistance-act

[2] M. Ralston, E. Schatz, J. Menken, F. X. Gómez-Olivé, and S. Tollman, "Who benefits—or does not—from South Africa's old age pension? Evidence from characteristics of rural pensioners and non-pensioners," International Journal of Environmental Research and Public Health, vol. 13, no. 1, Art. 85, 2016, doi: 10.3390/ijerph13010085.

[3] Statistics South Africa, Living Conditions Survey 2014/2015: Metadata, Statistical Release P0310. Pretoria, South Africa, 2017.

[4] Statistics South Africa, Living Conditions Survey: COICOP Codes. LCS 2014/2015 code list.

[5] Statistics South Africa, Living Conditions Survey 2014/2015: Household Questionnaire. Survey instrument.

[6] E. B. Godfrey et al., "Which eligible households get grants? Demographic correlates of receipt in South Africa," Development Southern Africa, vol. 33, no. 6, pp. 774–789, 2016, doi: 10.1080/0376835X.2016.1231059.

[7] T. Raniga and B. Simpson, "Poverty, HIV/AIDS and the old age pension in Bhambayi, KwaZulu-Natal, South Africa," Development Southern Africa, vol. 28, no. 1, pp. 75–85, 2011, doi: 10.1080/0376835X.2011.545171.

[8] M. Alloush, J. R. Bloem, and J. G. Malacarne, "Social protection amid a crisis: New evidence from South Africa's Older Person's Grant," The World Bank Economic Review, vol. 38, no. 2, pp. 371–393, 2024, doi: 10.1093/wber/lhad037.

[9] A. Case, V. Hosegood, and F. Lund, "The reach and impact of Child Support Grants: Evidence from KwaZulu-Natal," Development Southern Africa, vol. 22, no. 4, pp. 467–482, 2005, doi: 10.1080/03768350500322925.

[10] W. Zembe-Mkabile, T. Doherty, D. Sanders, and D. Jackson, "Child Support Grant access and receipt among 12-week-old infants in an urban township setting in South Africa," Global Health Action, vol. 7, no. 1, Art. 25310, 2014, doi: 10.3402/gha.v7.25310.

[11] J. Gazeaud, "Proxy means testing vulnerability to measurement errors?," The Journal of Development Studies, vol. 56, no. 11, pp. 2113–2133, 2020, doi: 10.1080/00220388.2020.1715942.

[12] D. P. Moynihan, P. Herd, and H. Harvey, "Administrative burden: Learning, psychological, and compliance costs in citizen-state interactions," Journal of Public Administration Research and Theory, vol. 25, no. 1, pp. 43–69, 2015, doi: 10.1093/jopart/muu009.

[13] C. J. Heinrich and R. Brill, "Stopped in the name of the law: Administrative burden and its implications for cash transfer program effectiveness," World Development, vol. 72, pp. 277–295, 2015, doi: 10.1016/j.worlddev.2015.03.015.

[14] A. Gibbs, L. Washington, N. Jama Shai, Y. Sikweyiya, and S. Willan, "Systematically excluded: Young women's experiences of accessing child support grants in South Africa," Global Public Health, vol. 13, no. 12, pp. 1820–1830, 2018, doi: 10.1080/17441692.2018.1449231.

[15] T. N. Majola and P. Maharaj, "The challenges associated with the Child Support Grant application process: A case of young mothers in a South African township," Journal of Social Service Research, vol. 51, nos. 5–6, pp. 1140–1150, 2025, doi: 10.1080/01488376.2025.2502004.

[16] S. Granlund, "They are not leaving without their money: Exploring encounters with the state through cash transfers in the rural Eastern Cape, South Africa," Southern African Journal of Social Work and Social Development, vol. 34, no. 1, Art. 9052, 2022, doi: 10.25159/2708-9355/9052.

[17] A. N. Vujovic, G. M. Jonathan, and S. Hacks, "Digitalisation and its impact on social protection programmes effectiveness: A case from South Africa," Procedia Computer Science, vol. 237, pp. 882–890, 2024, doi: 10.1016/j.procs.2024.05.177.

[18] M. T. Mabala, "Factors associated with social grant uptake in South Africa: A closer look at the Special COVID-19 Social Relief of Distress Grant among the youth aged 18–35 years," Master's research report, University of the Witwatersrand, Johannesburg, South Africa, 2023. Available: https://hdl.handle.net/10539/40003

[19] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, "Calibration: the Achilles heel of predictive analytics," BMC Medicine, vol. 17, Art. 230, 2019, doi: 10.1186/s12916-019-1466-7.

[20] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," PLOS ONE, vol. 10, no. 3, Art. e0118432, 2015, doi: 10.1371/journal.pone.0118432.

[21] R. D. Riley et al., "Minimum sample size for developing a multivariable prediction model: PART II—binary and time-to-event outcomes," Statistics in Medicine, vol. 38, no. 7, pp. 1276–1296, 2019, doi: 10.1002/sim.7992.

[22] D. R. Roberts et al., "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," Ecography, vol. 40, no. 8, pp. 913–929, 2017, doi: 10.1111/ecog.02881.

[23] G. S. Collins, J. B. Reitsma, D. G. Altman, and K. G. M. Moons, "Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): The TRIPOD statement," Annals of Internal Medicine, vol. 162, no. 1, pp. 55–63, 2015, doi: 10.7326/M14-0697.